

DW01_satrnd

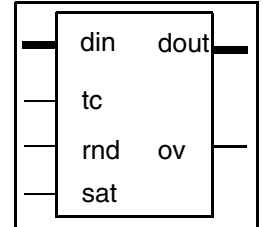
Arithmetic Saturation and Rounding Logic

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

Features and Benefits

- Parameterized word length
- Dynamically or statically configurable
- Arithmetic saturation (clipping) or wrap-around for MSB truncation
- Round to nearest logic for LSB truncation
- Signed and unsigned data operation

Revision History



Applications

- Digital Signal Processing (DSP)
- Graphics

Description

DW01_satrnd performs arithmetic, precision-handling rounding and saturation functions on its input bus *din*.

Table 1-1 Pin Description

Pin Name	Width	Direction	Function
din	<i>width</i> bits	Input	Input data
tc	1 bit	Input	Two's complement control ■ 0 = Unsigned ■ 1 = Signed
sat	1 bit	Input	Saturation enable ■ 0 = No saturation ■ 1 = Enable saturation
rnd	1 bit	Input	Rounding enable ■ 0 = No rounding ■ 1 = Enable rounding
ov	1 bit	Output	Overflow status
dout	<i>msb_out</i> – <i>lsb_out</i> + 1 bits	Output	Output data

Table 1-2 Parameter Description

Parameter	Values	Description
width	≥ 2 Default: 16	Word length of <code>din</code>
msb_out	$width-1 \geq msb_out > lsb_out$ Default: 15	<code>dout</code> MSB position after truncation of <code>din</code> MSBs
lsb_out	$msb_out > lsb_out \geq 0$ Default: 0	<code>dout</code> LSB position after truncation of <code>din</code> LSBs

Table 1-3 Synthesis Implementations

Implementation Name	Function	License Feature Required
str	Synthesis model	DesignWare

Table 1-4 Simulation Models

Model	Function
DW01.DW01_SATRND_CFG_SIM	Design unit name for VHDL simulation
dw/dw01/src/DW01_satrnd_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW01_satrnd.v	Verilog simulation model source code

The width of the `din` bus is set with the `width` parameter. The output bus, `dout`, is a subset of `din`. The width of `dout` is determined by the `msb_out` and `lsb_out` parameters such that `dout` equals `din(msb_out:lsb_out)`.

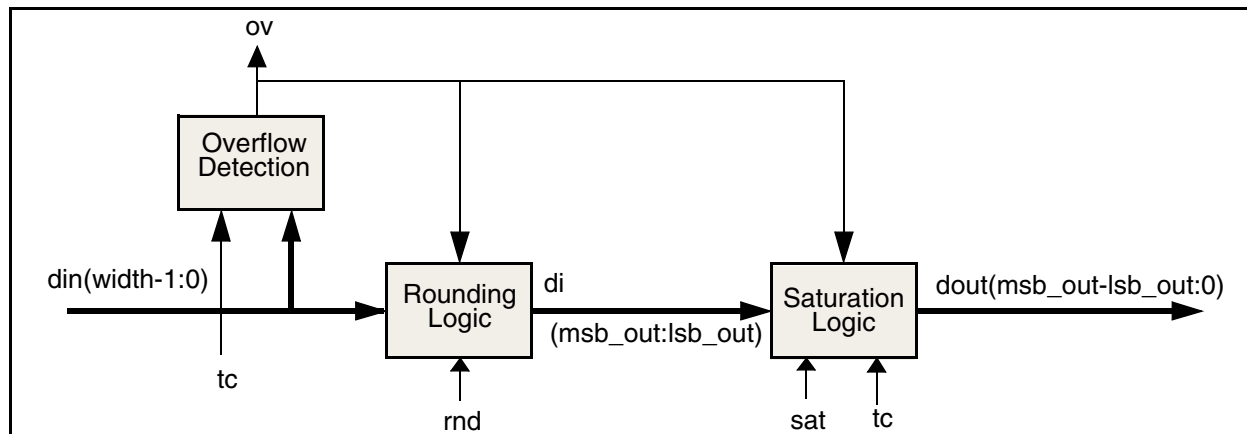
When the least significant bits `din(lsb_out-1:0)` are truncated, the round-to-nearest operation is performed if rounding is enabled (`rnd = 1`) and overflow has not occurred (`ov = 0`).

When the most significant bits `din(width-1:msb_out+1)` are truncated or rounding of a maximally positive value has occurred, data wrap-around may occur as a consequence of the finite data word length.

If saturation is enabled (`sat = 1`) and overflow has occurred (`ov = 1`), the output `dout` is saturated (or clipped) according to whether:

- The input data `din` is signed (`tc = 1`) or unsigned (`tc = 0`), and
- The sign of `din`.

Saturation always takes precedence over rounding. If overflow occurs and both saturation and rounding modes are enabled, the output `dout` is the saturation value.

Figure 1-1 Block Diagram

Related Topics

- [Math – Arithmetic Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_foundation_comp.all;

entity DW01_satrnd_inst is
  generic ( inst_width    : POSITIVE := 8;
            inst_msb_out  : NATURAL  := 7;
            inst_lsb_out  : NATURAL  := 1 );
  port ( inst_din   : in std_logic_vector(inst_width-1 downto 0);
        inst_tc    : in std_logic;
        inst_sat    : in std_logic;
        inst_rnd    : in std_logic;
        ov_inst     : out std_logic;
        dout_inst   : out std_logic_vector(inst_msb_out-inst_lsb_out downto 0)
        );
end DW01_satrnd_inst;

architecture inst of DW01_satrnd_inst is
begin

  -- Instance of DW01_satrnd
  U1 : DW01_satrnd
    generic map ( width => inst_width,    msb_out => inst_msb_out,
                  lsb_out => inst_lsb_out )
    port map ( din => inst_din,    tc => inst_tc,    sat => inst_sat,
              rnd => inst_rnd,    ov => ov_inst,    dout => dout_inst );
end inst;

-- pragma translate_off
configuration DW01_satrnd_inst_cfg_inst of DW01_satrnd_inst is
  for inst
    end for; -- inst
end DW01_satrnd_inst_cfg_inst;
-- pragma translate_on
```

HDL Usage Through Component Instantiation - Verilog

```
module DW01_satrnd_inst( inst_din, inst_tc, inst_sat, inst_rnd,
                        ov_inst, dout_inst );

    parameter width = 8;
    parameter msb_out = 7;
    parameter lsb_out = 1;

    input [width-1 : 0] inst_din;
    input inst_tc;
    input inst_sat;
    input inst_rnd;
    output ov_inst;
    output [msb_out-lsb_out : 0] dout_inst;

    // Instance of DW01_satrnd
    DW01_satrnd #(width, msb_out, lsb_out)
        U1 ( .din(inst_din),    .tc(inst_tc),    .sat(inst_sat),
            .rnd(inst_rnd),    .ov(ov_inst),    .dout(dout_inst) );

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes
January 2019	DWBB_201806.5	■ Updated example in “ HDL Usage Through Component Instantiation - VHDL ” on page 4 ■ Added this Revision History table and the document links on this page

Copyright Notice and Proprietary Information

© 2024 Synopsys, Inc. All rights reserved. This Synopsys software and all associated documentation are proprietary to Synopsys, Inc. and may only be used pursuant to the terms and conditions of a written license agreement with Synopsys, Inc. All other use, reproduction, modification, or distribution of the Synopsys software or the associated documentation is strictly prohibited.

Destination Control Statement

All technical data contained in this publication is subject to the export control laws of the United States of America. Disclosure to nationals of other countries contrary to United States law is prohibited. It is the reader's responsibility to determine the applicable regulations and to comply with them.

Disclaimer

SYNOPSYS, INC., AND ITS LICENSORS MAKE NO WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, WITH REGARD TO THIS MATERIAL, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE.

Trademarks

Synopsys and certain Synopsys product names are trademarks of Synopsys, as set forth at <https://www.synopsys.com/company/legal/trademarks-brands.html>.

All other product or company names may be trademarks of their respective owners.

Free and Open-Source Software Licensing Notices

If applicable, Free and Open-Source Software (FOSS) licensing notices are available in the product installation.

Third-Party Links

Any links to third-party websites included in this document are for your convenience only. Synopsys does not endorse and is not responsible for such websites and their practices, including privacy practices, availability, and content.

Synopsys, Inc.
www.synopsys.com

